FORM HS-07 · REV. 4 POST-STASIS REFRESHER · MODULE 7 OF 9 POST AT EVERY MACHINE SHOP AND ISOLATION POINT

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Health & Safety

LOCK IT OUT.

BEFORE YOU PUT ANY PART OF YOUR BODY INSIDE A MACHINE, EVERY SOURCE OF ENERGY GOES TO ZERO, UNDER YOUR OWN LOCK.

YOUR LOCK

ONE PERSON. ONE LOCK. ONE KEY.

Every worker who goes into a machine puts on **their own lock**. You hold the only key. Nobody else's lock protects you. Ten workers on one machine means ten locks.

Your tag shows your name, colonist ID, crew, date and the reason. A tag alone is a warning. **Only a lock stops the machine.**

Your personal lock is issued by Stasis Revival Unit with your kit and engraved with your colonist ID. Lost it? Report it to your foreman the same shift.

THE LOCKOUT SEQUENCE

OSHA 1910.147 · EVERY TIME, NO SHORTCUTS

LOCK OUT

1 **Prepare.** Read the machine's lockout procedure card. Know every energy source on it and where each isolation point is.

2 **Notify.** Tell the operator and everyone who works on or near the machine that it is going down and why.

3 **Shut down** using the normal stop. Never kill it with the emergency stop alone.

4 **Isolate every energy source.** Open the disconnect, close the valves, unplug, decouple. Pushing a control button is not isolation.

5 **Apply your lock and tag** to each isolation point. Use a hasp when more than one person locks out.

6 **Release or block stored energy.** Bleed pressure, drain capacitor banks, block raised parts, let flywheels stop, let hot parts cool.

7 **Verify zero energy.** Clear everyone away, then try to start it. Test electrical with a meter you have checked on a live source. Return controls to off.

8 **Do the work.**

RESTORE

1 **Tools and parts out.** Count them. Nothing left inside the machine.

2 **Guards back on** and secured.

3 **People clear.** Everyone accounted for and out of the danger zone.

4 **Notify** the operator and affected workers that power is coming back.

5 **Remove your own lock.** Only yours. Then the machine can be energized.

FIND EVERY ENERGY SOURCE

MOST MACHINES HAVE MORE THAN ONE

ELECTRICAL

Mains, generator feeds, control circuits. Isolate at the disconnect. Test before touch.

POWER CELLS & CAPACITOR BANKS

Stored charge stays after disconnect. Discharge through the rated grounding stick and verify zero on the meter. Some banks need minutes to bleed down.

MECHANICAL

Flywheels, springs, belts and gears still moving or under tension. Wait for full stop. Block or release springs.

HYDRAULIC

Pressure stays trapped in lines and accumulators. Bleed to zero. Hydraulic fluid injected under skin is a surgical emergency.

PNEUMATIC

Close the supply, then bleed the line. A charged air line can whip or fire a cylinder.

GRAVITY

Raised booms, rams, platforms, loads. Lower them or block them with rated blocks and pins. Never trust hydraulics to hold.

THERMAL

Ovens, presses, printer heads, pipes. Lock out the heat source and wait for safe touch temperature.

CHEMICAL

Feed lines for resins, gases, coolants. Close and lock valves, blind or disconnect the line, drain it.

MAGNETIC FIELD RIGS

Lifting magnets and field-forming rigs. A field rig can hold or fling steel after the switch is off until its coils are dumped. Lower the load first, dump the coils, verify zero field with the gauss meter.

GROUP LOCKOUT & SHIFT CHANGE

- **Big jobs use a lockbox.** The lead locks every isolation point with job locks, puts the keys in the yellow lockbox, and every worker puts a personal lock on the box.
- The box can't open until **the last worker removes their lock.**
- **Shift change:** the incoming worker puts their lock on **before** the outgoing worker takes theirs off. The machine is never unprotected.
- Walk the incoming crew through what is isolated and what is still live.

SOMEONE ELSE'S LOCK

You never cut, open or remove another worker's lock. The only exception is the formal procedure:

- The **foreman** confirms the worker is not at the machine.
- Every reasonable effort is made to **reach the worker**: radio, their sector channel, their barracks.
- The foreman removes it, logs it, and **tells the worker before they return to work.**

LOCK COLORS

DCW SITE STANDARD

RED · PERSONAL
Everyone's own lock. Engraved with your colonist ID. Only you hold the key.

YELLOW · JOB LOCK
Group lockout. Keys go in the lockbox.

BLUE · POWER ISOLATION
Electrical & Power division locks on substations and feeders. Their work, their release.

MACHINE GUARDING

FOR MACHINISTS AND FAB YARD CREWS

POINT OF OPERATION

Where the cut, press, punch or bend happens. Guards, light curtains or two-hand controls keep your hands out.

ROTATING PARTS

Shafts, spindles, couplings, chucks. They grab cloth, hair and gloves and pull you in.

NIP POINTS

Where two parts turn toward each other: belts and pulleys, gears, rollers. Never reach in to clear them.

FLYING CHIPS

Chips, sparks and broken tools. Safety glasses always, face shield on grinders, chip shields in place.

DO

- Roll or button sleeves tight. Tie back and cover long hair.
- Remove rings, watches, chains and lanyards before running lathes, mills and drill presses.
- Clamp the work. Never hold small parts by hand on a drill press.
- Use a brush or hook for chips. Stop the machine first.
- Report a missing or damaged guard and tag the machine out.

DON'T

- **No gloves** near rotating lathes, mills or drill presses. A glove gets caught and pulls in the whole hand.
- **Never leave the chuck key in the chuck.** It becomes a projectile at startup.
- Never reach around, over or under a guard.
- Never bypass, tape over or jumper an interlock or light curtain.
- Never clear a jam with the machine running.

CAUGHT IN

**STOP THE MACHINE.
THEN HELP.**

Reaching in to pull someone out of a running machine puts two people in it.

1. **Hit the emergency stop** or kill the disconnect.
2. **Ch 1: "Emergency, emergency, emergency. Caught-in,"** with sector-grid, deck and number hurt.
3. **Don't reverse the machine** to free them unless the medic or rescue says so. It can cause more injury.
4. **Control bleeding** with direct pressure. Apply a tourniquet above a limb wound that won't stop.
5. **Amputated part:** wrap in clean damp cloth, seal in a bag, keep the bag cool on ice. Don't put the part directly on ice.

STOP-WORK AUTHORITY

Missing guard, no lockout procedure, someone working inside a machine without their own lock: **stop the work** and call your foreman. No penalty, no questions.

WORKER ACKNOWLEDGEMENT

I have read this poster. I have my personal lock and tags, I know the lockout sequence, and I will never remove another worker's lock.

NAME
COLONIST ID
CREW / SECTOR
DATE

Adapted from Old Earth standards: OSHA 29 CFR 1910.147 (control of hazardous energy), OSHA 1910 Subpart O & 1910.212 (machine guarding), ANSI/ASSP Z244.1 (lockout, tagout and alternative methods), NFPA 70E (electrical safety in the workplace).